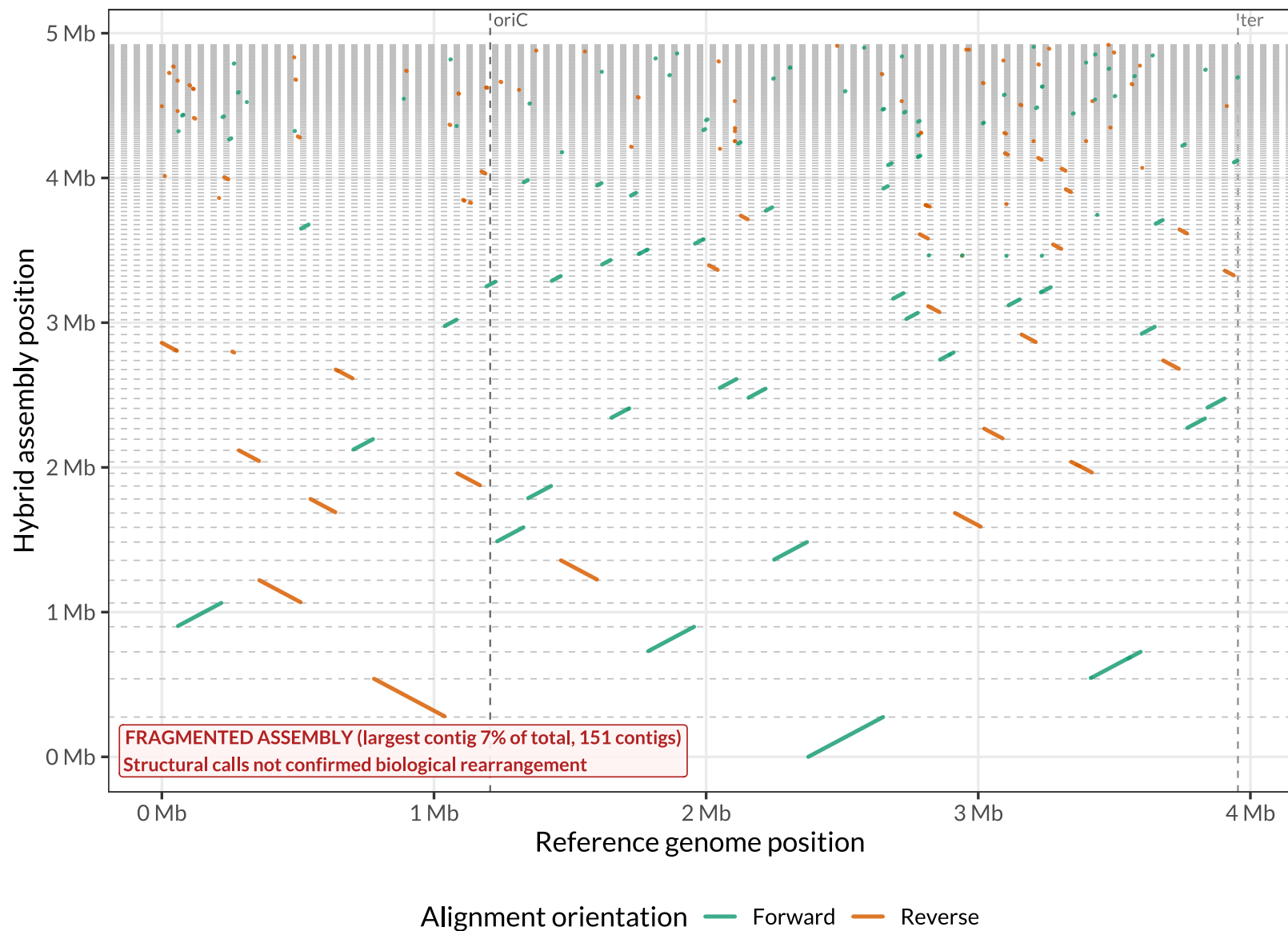


Whole-genome synteny: LAC-4 hybrid assembly vs. reference (FastANI = 99.9309%)



Strain LAC-4. Each segment is one MUMmer/NucDiff alignment block between the reference genome (x, Mb) and the Unicycler hybrid assembly (y, Mb); 151 assembly contig(s) are stacked with a cumulative offset (grey dashed boundaries, largest first). 202 alignment blocks shown, mean %identity (block-length-weighted) = 99.67%. Colored by alignment orientation: off-diagonal blocks indicate relocated/reshuffled sequence; anti-diagonal (Reverse) blocks indicate inversions. Dashed vertical lines mark oriC (position 0) and ter. CAUTION: this hybrid assembly is fragmented (largest contig = 6.7% of total length across 151 contigs) -- most structural differences from this comparison are likely assembly artifacts (contig-boundary effects) rather than confirmed biological rearrangement.